

RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences

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Pharmaceutical sponsors developing a drug for both the United States and the European Union must reconcile guidance issued independently by the FDA and the EMA. Where the two agencies require *substantively the same thing*, a sponsor can file once; where they *diverge*, a single trial design risks rejection in one region; where one agency is *silent* on a point the other regulates, the sponsor must infer obligations. Today this reconciliation is performed manually by regulatory-affairs experts. We introduce **cross-jurisdiction regulatory divergence detection**: given an FDA requirement and an EMA requirement on the same topic, classify their relationship as AGREE, DIVERGE, or SILENT. We release **RegDivergence-101**, a 101-pair expert-grounded benchmark (labels grounded in three peer-reviewed FDA/EMA comparison studies and primary FDA/EMA/ICH guidance text; inter-annotator $\kappa = 0.54$), and systematically characterize a four-method baseline hierarchy: lexical heuristic (0.556 macro-F1, 95% CI [0.458–0.650]), NLI cross-encoder (0.271), obligation-level Graph-RAG (0.698 [0.602–0.783]), and flat LLM judge / Claude Haiku (0.819 [0.736–0.900]). Three findings emerge: SILENT is semantically detectable but invisible to entailment-only formulations; pair-level obligation graphs improve over lexical methods but lose to flat-LLM context; and corpus-level graph construction is the correct architectural target for large-scale silent-detection.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Information systems** → **Document filtering**; • **Applied computing** → **Health informatics**.

Additional Key Words and Phrases: large language models, LLM benchmark, regulatory NLP, AI for compliance, FDA/EMA divergence, contradiction detection, Graph-RAG, retrieval-augmented generation, generative AI, life sciences, AGREE/DIVERGE/SILENT classification

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1 Introduction

Bringing a medicine to market in both the US and EU requires satisfying two regulators that publish guidance separately and, frequently, *differently*. The canonical example is the choice of primary **estimand** under ICH E9(R1): for the same trial, FDA guidance favors the *treatment-policy* strategy while EMA guidance favors the *hypothetical* strategy. Other documented divergences span pediatric extrapolation (ICH E11A), endpoint definitions in ulcerative colitis, gene-therapy long-term follow-up duration, and the regulatory pathway for companion diagnostics. A sponsor who misses one of these divergences during protocol design can lose a year or more re-running a study.

Reconciling the two corpora is currently a manual, expert task. Yet the NLP building blocks for automating it all exist—document-level NLI (ContractNLI [8], DocNLI [15]), legal contradiction detection (LegalWiz [10], CLAUSE [12]), and Graph-RAG (Microsoft GraphRAG [5], GraphCompliance [4], RAGulating Compliance [1]). **No prior work combines them to detect divergence between two parallel regulatory corpora from different jurisdictions.** All existing contradiction work is *intra-document* (within one contract) or *single-jurisdiction* (contract-vs-statute).

This paper makes three contributions:

- (1) **Task formulation.** Cross-jurisdiction regulatory divergence detection as a three-way classification (AGREE / DIVERGE / SILENT) over aligned FDA/EMA requirement pairs, grounded in published expert comparisons.
- (2) **Benchmark. RegDivergence-101**, 101 expert-grounded pairs (59 from peer-reviewed FDA/EMA comparison studies, 42 derived from primary FDA/EMA/ICH guidance) spanning 13 topic areas, with inter-annotator agreement ($\kappa = 0.54$) and bootstrap confidence intervals.
- (3) **Baseline hierarchy with three findings.** Four methods systematically compared, yielding empirical insights on the task’s structure and the path to a corpus-level solution.

2 Related Work

Document-level NLI. ContractNLI [8] introduced document-level entailment/contradiction/not-mentioned classification with evidence extraction, and showed that contradiction detection lags entailment, especially under “negation by exception.” DocNLI [15] extended NLI to full documents.

Legal contradiction detection (2025–2026). LegalWiz [10] uses a multi-agent generation framework for legal contradiction detection but is confined to single-document regimes. Better Call CLAUSE [12] benchmarks 7,500+ perturbed contracts across ten anomaly categories and finds that LLMs miss subtle errors and struggle to *justify* them legally. Contradiction Detection in RAG Systems [6] uses LLMs as context validators. Two recent RAG-faithfulness studies bear directly on our LLM-judge and graph-RAG results: Chen et al. [3] probe whether a RAG system can recognize when retrieved context is incorrect, and Qian et al. [11] show that the surface relevance of retrieved evidence does not guarantee it *warrants* a claim—the same failure mode behind our Finding 2, where modal-register similarity (RECOMMENDED vs. MANDATORY) misleads the pair-level graph.

Graph-RAG and regulatory compliance. Microsoft GraphRAG [5] builds entity graphs with community summaries for local and global queries. GraphCompliance [4] aligns *policy graphs* (regulatory requirements) with *context graphs* (organizational facts) for GDPR; it explicitly lists “handling conflicting

regulatory interpretations” as open. RAGulating Compliance [1] builds an ontology-free regulatory knowledge graph for traceable QA.

Gap. Contradiction detection is intra-document or single-jurisdiction; Graph-RAG targets single-corpus QA or single-jurisdiction compliance; FDA/EMA divergence is documented manually but never automated. Our task sits precisely in this gap.

3 Task and Dataset

3.1 Task Definition

For a topic t , let f_t be an FDA requirement statement and e_t the corresponding EMA statement. The label y_t is:

- **AGREE** — both jurisdictions require substantively the same thing.
- **DIVERGE** — the requirements directly conflict (one mandates what the other forbids, qualifies, or sets a different threshold for).
- **SILENT** — one jurisdiction does not address what the other regulates.

3.2 Dataset Construction

We seed pairs from documented FDA/EMA divergences in three open-access expert-panel reviews: a *J. Crohn’s & Colitis* 2025 expert comparison of FDA and EMA ulcerative colitis guidance (40 pairs) [14], a PMC study comparing EMA and FDA psychiatric drug-trial guidelines (10 pairs) [2], and a PMC comparison of US and EU radiopharmaceutical nonclinical requirements (9 pairs) [13]; supplemented by 42 pairs derived from primary guidance text (ICH E9(R1), E11A [7]; FDA and EMA guidance). Each pair records `{id, topic, source, fda_text, ema_text, label}`. The combined release contains **101 pairs**: 38 AGREE, 40 DIVERGE, 23 SILENT, across 13 topic areas. We report macro-F1 to weight all classes equally.

Inter-annotator agreement. To validate the labeling schema, a second independent annotator (Claude Haiku with a deliberately different prompt and no access to the original labels) re-labeled a stratified sample of 36 pairs (12 per class). Observed agreement: 0.694; Cohen’s $\kappa = 0.542$ (moderate, Landis & Koch 1977 [9]). AGREE was the most reliable class (83% agreement); SILENT the hardest (58%, mainly confused with DIVERGE). We disclose that both annotation passes are LLM-assisted; the expert-panel source papers provide the citable human-expert anchor. The moderate κ is below the conventional 0.60 threshold and reflects genuine task difficulty—SILENT requires determining *absence* of regulation, which is inherently more ambiguous than detecting presence of conflict. We treat $\kappa = 0.542$ as a characterization of task difficulty rather than a quality failure, and recommend that a full human-expert IAA study be conducted for the camera-ready benchmark release.

4 Baselines

We evaluate four methods under a shared reporting framework: **stratified 5-fold splits** averaged over 5 random seeds, with **bootstrap 95% CI** (1,000 resamples). However, the protocol differs by method class. The lexical heuristic and NLI cross-encoder have tunable thresholds and use **nested** CV (inner folds tune thresholds, outer folds report). The Graph-RAG classifier and LLM judge have no tunable hyperparameters:

the Graph-RAG classifier is reported over the same outer folds, and the LLM judge is scored directly on all 101 pairs. CIs for both LLM methods are bootstrap-only.

4.1 Lexical Heuristic

No neural model, no API—fully offline and deterministic. Combines TF-IDF cosine similarity with five conflict signals: negation asymmetry, divergence cue pairs (e.g., *single-arm* vs. *randomized*), modal-strength asymmetry (*must/shall* vs. *may/encouraged*), numeric-threshold mismatch, and one-sided specificity (concrete number vs. case-by-case deferral). A shared ICH citation is a strong AGREE signal.

4.2 NLI Cross-Encoder

`typeform/distilbert-base-uncased-mnli`, run in both directions; labels derived from contradiction/neutral/entailment probabilities with CV-tuned thresholds. We use DistilBERT-MNLI as a representative off-the-shelf NLI system. The structural SILENT gap finding (§7, Finding 1) is model-agnostic: any NLI model lacking an explicit SILENT output class will map absence-of-regulation to *neutral*, collapsing SILENT recall. Substituting DeBERTa-v3-large or a modern cross-encoder would improve AGREE/DIVERGE F1 but would not resolve the SILENT structural gap without task-specific supervision.

4.3 Graph-RAG Pair-Level Classifier

For each text, Claude Haiku extracts a policy triplet $\langle \text{subject}, \text{obligation_level}, \text{requirement}, \text{conditions} \rangle$ where $\text{obligation_level} \in \{\text{MANDATORY}, \text{PROHIBITED}, \text{RECOMMENDED}, \text{PERMITTED}, \text{SILENT}\}$. Only $\text{MANDATORY} \leftrightarrow \text{PROHIBITED}$ is a hard-coded DIVERGE rule; all other pairs are escalated to a second LLM call that reasons over the *obligation structure* of the aligned node pair, with explicit calibration that modal-register differences (RECOMMENDED vs. MANDATORY) do not automatically imply DIVERGE.

Evaluation note. The graph classifier was evaluated on the same 101 pairs as all other methods, using the same stratified nested CV outer folds for reporting. No hyperparameter tuning was performed (the $\text{MANDATORY} \leftrightarrow \text{PROHIBITED}$ rule and the LLM prompt were fixed before evaluation began). Bootstrap CI is over the same held-out fold predictions as the other methods.

4.4 LLM Judge (Flat Pairwise)

Claude Haiku with explicit AGREE / DIVERGE / SILENT definitions, prompted once per pair with both texts. No graph, no retrieval. Full prompt in Appendix A.

5 Results

Table 1 reports macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for all four methods.

Point estimates increase monotonically from lexical to Graph-RAG to the LLM judge. At $n = 101$ the marginal bootstrap CIs partially overlap, so we report these as a consistent ranking trend rather than a claim of statistical significance; a paired significance test on shared held-out predictions is left to future work.

Table 1. Macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for the four baselines ($n = 101$ pairs).

Method	Macro-F1 [95% CI]	Acc	AGREE F1 [CI]	DIVERGE F1 [CI]	SILENT F1 [CI]
Lexical heuristic ¹	0.556 [0.458–0.650]	0.586	0.756 [0.648–0.849]	0.538 [0.400–0.660]	0.361 [0.167–0.545]
NLI cross-encoder ²	0.271 [0.181–0.354]	0.277	0.303 [0.118–0.481]	0.416 [0.278–0.545]	0.079 [0.000–0.177]
Graph-RAG pair-level ³	0.698 [0.602–0.783]	0.703	0.766 [0.646–0.860]	0.691 [0.554–0.806]	0.628 [0.468–0.764]
LLM judge (Haiku)⁴	0.819 [0.736–0.900]	0.832	0.884 [0.800–0.951]	0.830 [0.737–0.911]	0.743 [0.579–0.878]

¹Nested 5-fold CV, 5 seeds; thresholds tuned on inner folds only. ²Same CV protocol, thresholds tuned on inner folds. ³No tunable hyperparameters; bootstrap CI only. ⁴claude-haiku-4-5-20251001, default temperature; bootstrap CI only. Estimated cost: ~\$0.10 for all 101 pairs.

6 Error Analysis: The Lexical Ceiling

The lexical heuristic misclassifies roughly 42 of the 101 pairs (accuracy 0.586). The illustrative errors below share a signature: **high lexical overlap, opposed regulatory stance**.

- **adaptive-01 (DIVERGE → missed):** FDA “does *not require* detailed pre-specification” vs. EMA “should be *fully prespecified*”—near-identical vocabulary, opposite obligation.
- **uc-01:** Single-arm Mayo-score endpoint vs. co-primary modified-Mayo—same terms, different structural requirement.
- **gene-01:** FDA recommends 15 years; EMA defers to case-by-case—same topic, different specificity.

A bag-of-words model cannot resolve these; they require representing *what each requirement asserts about which entity*.

7 Discussion: What the Baseline Hierarchy Reveals

Finding 1 — SILENT is semantically detectable but invisible to NLI framing. SILENT F1 = 0.079 under NLI (no native SILENT class) vs. 0.743 under the LLM judge (explicit definition). This is not a model-capacity effect—it is structural. The SILENT class requires a formulation that can express absence; NLI entailment cannot.

Finding 2 — Pair-level obligation graphs improve over lexical (+0.14 F1) but lose to flat LLM (−0.12 F1). The graph extracts and compares obligation levels (MANDATORY / RECOMMENDED / etc.), which helps distinguish clear stance differences from similar-sounding text. But it loses the full-text context that the flat LLM uses to separate *same-requirement* / *different-modal-register* (AGREE) from *same-topic* / *different-threshold* (DIVERGE). Pairs like gene-01 (FDA 15 years vs. EMA case-by-case) both extract as RECOMMENDED and the graph calls them AGREE; the flat LLM correctly identifies the threshold conflict from the text. This is precisely the failure mode identified by Qian et al. [11]: surface relevance does not imply evidential force.

Finding 3 — Two distinct ceilings for two different problem scopes. The *pair-level ceiling* is the flat LLM judge (0.819): given two aligned sentences and explicit definitions, an LLM nearly saturates the task. The *corpus-level ceiling* is the domain of the graph method: “is EMA silent on this topic *anywhere* across its guideline corpus?” is a query a pairwise LLM cannot make but a corpus-level policy graph can.

These are different tasks with different upper bounds. This paper measures and establishes the pair-level ceiling; §8 motivates the corpus-level architecture as the natural next contribution.

8 Future Work: Corpus-Level Policy Graph

The pair-level experiments establish that the correct target for a genuine Graph-RAG contribution is **corpus-level** silent detection and divergence discovery—not re-ranking an already-aligned pair, but *finding* which FDA requirements have no EMA counterpart and vice versa.

The proposed architecture:

- (1) **Per-jurisdiction policy-graph construction**—index all FDA guidance documents and all EMA guidelines; extract requirement triplets at corpus scale.
- (2) **Cross-jurisdiction alignment**—entity normalization + embedding similarity to link corresponding requirement nodes across the two graphs.
- (3) **Divergence and silence discovery**—for each aligned node pair: DIVERGE if obligation structures conflict; SILENT if a concept node exists in one graph but has no aligned counterpart in the other.
- (4) **Explanation generation**—natural-language summaries for regulatory-affairs specialists.

Key open questions: (a) does a corpus-level graph recover SILENT cases that are undetectable pairwise? (b) what is the alignment precision when FDA and EMA use different terminology for the same concept? (c) can the graph be kept current as guidance is updated?

9 Limitations

Benchmark scale. 101 pairs from three source documents introduces domain skew (40/101 from one UC comparison paper) and limits statistical power. The CI widths (≈ 0.09 – 0.17 for most methods) reflect this. Results should be treated as directional.

Inter-annotator agreement. $\kappa = 0.542$ is moderate. Both annotation passes are LLM-assisted; no independent human-expert double annotation was performed. The expert-panel source papers provide the primary validity anchor, but a full human IAA study is needed for a final benchmark release.

Data contamination. Claude Haiku may have been exposed to FDA/EMA guidance and the published expert-panel comparison papers during pretraining. To probe this, we split performance by source: on the 42 *primary guidance pairs* (low contamination risk) Haiku achieves AGREE F1 = 0.941 and DIVERGE F1 = 0.968; on the 59 *expert-comparison pairs* (higher risk) AGREE F1 = 0.769 and DIVERGE F1 = 0.745. Performance is *not* higher on the potentially-contaminated split. The macro-F1 gap (0.636 vs. 0.765) is fully explained by SILENT support ($n = 2$ on primary vs. $n = 21$ on expert pairs), not memorization. Future work should evaluate on guidance issued after the model’s knowledge cutoff.

SILENT operationalization. Our SILENT pairs use a placeholder on the absent side, which may not reflect real retrieval conditions where a system must determine absence from a full corpus. The pair-level SILENT task is a proxy; §8 describes the correct corpus-level formulation.

10 Conclusion

We introduced cross-jurisdiction regulatory divergence detection, released **RegDivergence-101**, a 101-pair expert-grounded benchmark ($\kappa = 0.54$), and characterized a four-method baseline hierarchy with bootstrap CIs. Three empirical findings: SILENT requires explicit absence-aware formulation (NLI F1 0.08 $\rightarrow$ LLM F1 0.74); obligation-level graph structure improves over lexical heuristics (+0.14 F1) but loses to flat LLM context (−0.12 F1); and the graph method’s correct target is corpus-level construction, where it uniquely enables silence discovery at scale. The flat LLM judge (0.819) is the pair-level ceiling; beating it requires the corpus-level architecture proposed in §8.

Acknowledgments

This work was conducted independently. Benchmark pairs were derived from publicly available FDA/EMA guidance documents and open-access peer-reviewed comparison studies. LLM API calls used the Anthropic API (Claude Haiku, `claude-haiku-4-5-20251001`).

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A LLM Judge Prompt

System:

You are a regulatory-affairs expert comparing FDA and EMA guidance documents. Given an FDA requirement and an EMA requirement on the same topic, classify their relationship.

Respond with ONLY a JSON object in this exact format:

```
{"label": "<AGREE|DIVERGE|SILENT>", "reason": "<one sentence>"}
```

Definitions:

- AGREE: both jurisdictions require substantively the same thing
- DIVERGE: the requirements directly conflict (one mandates what the other forbids or softens)
- SILENT: one jurisdiction does not address what the other regulates

User:

FDA requirement:

```
{fda_text}
```

EMA requirement:

```
{ema_text}
```

Classify the relationship.

Model: claude-haiku-4-5-20251001. Temperature: default (1.0). Max tokens: 100. Results cached; each pair scored exactly once.